

1. What is Virtualization?

Virtualization is a technology that allows a single physical computer (hardware) to be divided into multiple **virtual machines (VMs)**. Each VM behaves like an independent computer with its own operating system, CPU, memory (RAM), storage, and network resources.

In simple terms, virtualization enables **one physical machine to run multiple isolated operating systems simultaneously**.

2. What is a Hypervisor?

A **hypervisor** is software (or firmware) that **creates, manages, and runs virtual machines (VMs)**.

It sits between the **physical hardware** and the **virtual machines** and is responsible for:

- Allocating hardware resources (CPU, RAM, storage, network)
- Ensuring isolation between VMs
- Managing VM lifecycle (start, stop, pause, delete)

Basic flow:

Physical Hardware



Hypervisor



Multiple Virtual Machines (VMs)

3. How Does a Hypervisor Work?

A hypervisor works by **abstracting physical hardware** and presenting it as virtual hardware to each VM.

Key functions:

1. Hardware Resource Sharing

- The hypervisor shares physical resources (CPU, RAM, disk, network) among multiple VMs.
- Example: VirtualBox uses your host machine's hardware and allocates portions of it to each VM.

2. Virtual Hardware Creation

- Each VM gets a **separate set of virtual components**:
 - Virtual CPU
 - Virtual RAM

- Virtual disk
- Virtual network interface

3. Isolation

- VMs are fully isolated from:
 - The host operating system
 - Other virtual machines
- A crash or security issue in one VM does **not affect** others.

4. Instruction Handling

- The hypervisor intercepts and manages privileged CPU instructions to ensure safe execution.
- Modern CPUs (Intel VT-x, AMD-V) provide hardware support for efficient virtualization.

4. Real-World Example

Assume you have a physical device with:

- **16 GB RAM**
- **1 TB storage**
- **1 CPU (multi-core)**

You install a hypervisor on this device.

Now you create a VM with:

- **4 GB RAM**
- **100 GB storage**
- **2 virtual CPU cores**

What happens:

- The hypervisor allocates 4 GB RAM and 100 GB disk from the physical machine.
- The VM behaves like a **separate computer**.
- You can install an operating system (e.g., Linux or Windows) inside that VM.
- You can create multiple such VMs as long as physical resources are available.

Effectively, each VM is a **“small computer inside your main computer.”**

5. Types of Hypervisors

Type 1 Hypervisor (Bare Metal)

- Installed **directly on physical hardware**
- No **host operating system** in between
- Offers better performance and security
- Commonly used in **cloud computing and enterprise data centers**

Examples:

- **VMware ESXi**
- Microsoft Hyper-V (bare-metal mode)
- Xen
- KVM (Linux-based)

Architecture:

Physical Hardware (AWS Server)



AWS Nitro Hypervisor (Type 1)



Multiple EC2 Instances (VMs)

Type 2 Hypervisor (Hosted)

- Installed **on top of a host operating system**
- Easier to use but slightly less efficient
- Commonly used for **development, testing, and learning**

Examples:

- **Oracle VirtualBox**
- **VMware Workstation**
- VMware Fusion
- Parallels Desktop

Architecture:

Hardware



Host Operating System



Type 2 Hypervisor



Virtual Machines

6. Summary

- **Virtualization** allows multiple operating systems to run on one physical machine.
- A **hypervisor** is the core software that enables virtualization.
- Hypervisors allocate hardware resources, create virtual hardware, and ensure isolation.
- **Type 1 hypervisors** run directly on hardware and are used in production environments.
- **Type 2 hypervisors** run on a host OS and are ideal for personal or development use.

Real-World Example: AWS as a Type 1 (Bare-Metal) Hypervisor

In **Amazon Web Services (AWS)**, virtualization is implemented using a **Type 1 (bare-metal) hypervisor**, meaning the hypervisor runs **directly on the physical server hardware** inside AWS data centers.

How AWS Uses Type 1 Hypervisors

1. **Physical Server (Bare Metal)**
 - High-performance hardware (CPU, RAM, SSD/NVMe, network cards).
 - No traditional host operating system is exposed to customers.
2. **AWS Hypervisor (Nitro System)**
 - AWS uses the **Nitro Hypervisor**, a lightweight bare-metal hypervisor.
 - It directly controls hardware resources.
 - Offloads networking, storage, and security to dedicated Nitro hardware cards.
3. **Virtual Machines (EC2 Instances)**
 - Each EC2 instance is a virtual machine.
 - Customers choose CPU, RAM, storage, and OS.
 - Instances are fully isolated from each other.

1. What is Cloud Computing?

Cloud computing is the on-demand delivery of computing resources—such as servers, storage, databases, networking, software, analytics, and intelligence—over the internet, with **pay-as-you-go pricing**.

Instead of buying and maintaining physical data centers and servers, organizations **rent resources from cloud providers** and scale them as needed.

Simple Metaphor (to remember)

Cloud = Electricity Grid

- You do not build your own power plant.
- You consume electricity when needed.
- You pay only for what you use.

Real-World Example

- Netflix uses AWS to stream videos globally.
- Startups host applications on cloud instead of buying servers.
- Companies store files on Google Drive instead of local disks.

2. Types of Cloud Computing (Service Models)

These describe **what level of control you get**.

1. IaaS – Infrastructure as a Service

Definition:

Provides basic infrastructure such as virtual machines, storage, and networks. You manage the OS and applications.

You manage:

OS, middleware, runtime, applications, data

Provider manages:

Physical servers, storage, networking, virtualization

Examples:

- AWS EC2
- Azure Virtual Machines

- Google Compute Engine

Real-World Example:

A startup launches a website by renting virtual servers instead of buying hardware.

Metaphor (Easy to Memorize):

IaaS = Renting an empty house

- You get walls, electricity, water.
 - You bring furniture and live your way.
-

2. PaaS – Platform as a Service

Definition:

Provides a platform to build and deploy applications without worrying about infrastructure or OS management.

You manage:

Applications and data

Provider manages:

Infrastructure, OS, runtime, middleware

Examples:

- AWS Elastic Beanstalk
- Google App Engine
- Azure App Service
- Heroku

Real-World Example:

Developers upload code, and the platform automatically handles scaling and server management.

Metaphor:

PaaS = Renting a furnished apartment

- Furniture and utilities are ready.
 - You just live (write code).
-

3. SaaS – Software as a Service

Definition:

Fully managed software delivered over the internet.

You manage:

Only usage and data

Provider manages:

Everything

Examples:

- Gmail
- Google Docs
- Salesforce
- Microsoft 365

Real-World Example:

Using Gmail without installing or maintaining an email server.

Metaphor:

SaaS = Hotel stay

- Everything is managed.
 - You just use the service.
-

3. Cloud Deployment Models

These define **where the cloud is hosted and who can access it**.

1. Public Cloud

Definition:

Resources are owned and operated by a third-party cloud provider and shared among multiple customers.

Examples:

- AWS
- Microsoft Azure
- Google Cloud

Real-World Example:

Netflix running on AWS.

Metaphor:

Public Cloud = Public transport

- Shared resources
 - Cost-effective
 - Scalable
-

2. Private Cloud

Definition:

Cloud infrastructure dedicated to a single organization.

Examples:

- On-premise OpenStack
- VMware private cloud

Real-World Example:

Banks hosting sensitive data in their own private cloud.

Metaphor:

Private Cloud = Personal car

- Full control
 - Higher cost
 - More security
-

3. Hybrid Cloud

Definition:

Combination of public and private clouds with data and applications shared between them.

Examples:

- AWS + on-premise data center
- Azure + private servers

Real-World Example:

Sensitive customer data stored privately, while web apps run on public cloud.

Metaphor:

Hybrid Cloud = Car + public transport

- Flexibility
 - Balance of cost and control
-

4. Common Cloud Services Offered by Providers

1. Compute Services

Virtual machines and processing power.

- AWS EC2
- Azure VM

Metaphor: Engine of a car

2. Storage Services

Object, block, and file storage.

- AWS S3, EBS
- Azure Blob Storage

Metaphor: Warehouse

3. Database Services

Managed relational and NoSQL databases.

- Amazon RDS, DynamoDB
- Azure SQL

Metaphor: Digital filing cabinet

4. Networking Services

Virtual networks, load balancers, DNS.

- VPC, Load Balancer

Metaphor: Roads and traffic signals

5. Container Services

Run containerized applications.

- Kubernetes (EKS, AKS)
- Docker

Metaphor: Shipping containers

6. Serverless Computing

Run code without managing servers.

- AWS Lambda
- Azure Functions

Metaphor: Taxi service (pay per ride)

7. Machine Learning Services

Build and deploy ML models.

- AWS SageMaker
- Google Vertex AI

Metaphor: Smart assistant

8. Identity and Access Management (IAM)

Control who can access what.

- AWS IAM
- Azure AD

Metaphor: Security guard and ID cards

9. Monitoring and Logging

Track performance and logs.

- CloudWatch
- Azure Monitor

Metaphor: CCTV cameras

10. Content Delivery Network (CDN)

Deliver content faster worldwide.

- CloudFront
- Azure CDN

Metaphor: Nearby delivery hubs

11. Backup & Disaster Recovery

Data protection and recovery.

- AWS Backup
- Azure Site Recovery

Metaphor: Insurance policy

12. Security and Compliance Tools

Encryption, threat detection, audits.

- AWS Shield, GuardDuty

Metaphor: Alarm system

13. Virtual Private Cloud (VPC)

Isolated private network in the cloud.

- AWS VPC
- Azure VNet

Metaphor: Gated community

14. Data Analytics Services

Big data processing and analytics.

- AWS Athena, Redshift
- Google BigQuery

Metaphor: Business intelligence dashboard

15. API Management

Create, publish, and secure APIs.

- AWS API Gateway
- Azure API Management

Metaphor: Reception desk for services

Final Memory Trick (One Line)

Cloud computing is like running a business in a smart city where infrastructure, security, power, and services are rented, scalable, and managed for you—so you focus only on growth.